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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEFSC201 – QUESTIONED DOCUMENT EXAMINATION
(2024 ADMISSION ONWARDS)

Part A- 1 mark question

1. A qualified examiner must possess academic background in forensic science and specific training in questioned document analysis for reliability.
2. Magnifying lens / Stereomicroscope (or any relevant)
   These tools help in visual inspection to observe pen pressure, alignment, erasures, and indented writings.
3. Exemplar -Exemplar or standard writing is a known sample of handwriting used for comparison.
4. Law established through judicial decisions- Case law forms legal precedents guiding future judgments and forensic practices.
5. Scientific analysis of documents to determine authenticity and authorship/ It involves handwriting, typewriting, ink, paper, and signature analysis.
6. Position of writing relative to baseline- It indicates personality traits and helps in identifying individual writing habits.
7. Microprinting- Used to prevent counterfeiting and can be observed only under magnification.
8. Using chromatography or spectroscopic methods- TLC or HPLC helps separate ink components for identification.
9. Questioned and standard documents produced before the court- Any document authenticated and relevant to a case becomes legal evidence.
10. Brain (Motor cortex)- The central nervous system regulates muscle movement involved in writing.

PART B – Short Answer Questions (Any 5 × 3 = 15 Marks)

11. Digital enhancement helps to clarify faint, erased, or overwritten writings. Image processing software (like Adobe Photoshop or VSC systems) is used to adjust brightness, contrast, and sharpness. Infrared or ultraviolet filters can reveal writings invisible to the naked eye. Enhancements also help in photographing and documenting questioned areas for court presentation. **(Mention of image enhancement and software – 1 mark Infrared/UV technique – 1 mark, Purpose and forensic value – 1 mark)**
12. Documents must be handled with utmost care to prevent contamination or alteration. Examiners should use gloves or forceps, avoid folding, writing, or making marks on the document. Each exhibit should be packed in a clean, labelled cover and stored in dry conditions. Moisture, sunlight, and adhesives should be avoided to preserve integrity.(Precautions listed (3–4 points) – 2 marks, Emphasis on preservation and legal chain – 1 mark)
13. Disguised writing shows intentional modifications to conceal identity. Features include altered slant, inconsistent size and spacing, unnatural pen lifts, tremors, irregular pressure, and abnormal letter forms. Despite efforts, subconscious writing traits like rhythm and connecting strokes may persist, helping experts detect disguise. (Identification of altered characteristics 2 marks, Mention of persistent natural features – 1 mark)

14. Inkjet printers deposit liquid ink droplets forming visible dot patterns under magnification, often showing edge feathering on paper. Laser printers use toner powder fused to the paper by heat, producing smooth, uniform, and slightly raised prints. Toner may be removable with adhesive tape, while inkjet prints soak into fibers. (Printing principle (ink vs toner) – 1 mark, Physical difference (dot pattern, texture) – 1 mark, Forensic identification aspect – 1 mark)
15. Addition occurs when extra words, digits, or lines are inserted after writing (e.g., adding digits to cheque amount). Substitution involves replacing original text with new content, often by erasure or overwriting (e.g., changing date or name). Both are detected by magnification, oblique lighting, and chemical analysis of inks.
    (Definition of both – 2 marks, Example and detection – 1 mark)
16. Standard samples (exemplars) are essential for comparison to identify an author. They represent known handwriting under natural conditions, helping examiners assess similarities and differences in questioned documents. Without authentic exemplars, scientific comparison is unreliable. (Definition – 1 mark, Purpose – 1 mark, Importance in examination – 1 mark)
17. Magnification reveals disturbances in paper fibers, abrasions, and trace residues left after erasure. Under magnified view, erased regions appear rough, discolored, or uneven in texture. Magnifiers and stereomicroscopes also help differentiate chemical and mechanical erasures by observing paper surface details. (Description of magnification – 1 mark, Detection of physical features – 1 mark, Forensic interpretation – 1 mark)

PART C – Short Essay Questions (Any 5 × 6 = 30 Marks)

18. Forgery refers to the false making or alteration of a document with intent to deceive.
    Types include:
    Simulated (Imitated) forgery – imitation of genuine handwriting; detected by hesitation, tremors, inconsistent pressure.
    Traced forgery – copying over genuine writing using carbon or tracing paper; identified by slow movement, uniform thickness.
    Freehand forgery – forged without model; lacks genuine rhythm.
    Electronic forgery – use of scanners or digital methods to reproduce signatures.
    Detection involves examination of pen movement, line quality, pressure pattern, and microscopic analysis. Instruments like VSC and ESDA aid in confirming forgeries.
    Definition – 1
    Types (at least three) – 3
    Detection methods and examples – 2
19. A forensic report provides scientific findings and expert opinion for court.
    Structure includes:
    Heading and case details – Name, case number, date, authority.
    Items received – Exhibit list.
    Purpose of examination – Question framed by the investigating officer.
    Methodology – Scientific techniques used.
    Observations and results – Factual findings.
    Opinion / Conclusion – Expert's interpretation.
    Signature and seal – Authentication of report.
    Each section ensures clarity, transparency, and admissibility in legal proceedings.
    Marks:

Proper sequence of sections – 3 Significance and clarity – 3

20. Exemplars are known handwriting samples used for comparison.
    Types:
    Collected exemplars – Naturally written prior to investigation (letters, diaries, bank slips).

Requested exemplars – Written under supervision during investigation; ensures control and consistency.
Informal exemplars – Everyday writings collected from casual materials.
Collected exemplars show natural variation; requested ones are useful for specific text reproduction. Both are crucial for reliable authorship determination. (Definition – 1 Types (explained) – 4, Example – 1)

21. Handwriting has unique features reflecting motor coordination and habitual patterns. Examiners study characteristics like slant, size, alignment, spacing, connecting strokes, loops, pressure, and rhythm. Even in anonymous writings, subconscious traits persist. Comparison with known exemplars helps identify the probable writer using scientific reasoning and statistical evaluation. (**Marks:** List of characteristics – 3, Explanation of persistence – 2 Role in identification – 1)
22. Paper chromatography separates ink or dye mixtures.
    Steps: Sample preparation – Ink spot placed on baseline of filter paper, **Solvent preparation** – Suitable mobile phase (e.g., water, ethanol), **Development** – Paper suspended in solvent; capillary action separates components, **Drying and visualization** – Dry paper; visualize spots under UV or iodine chamber, **Interpretation** – Calculate Rf values; compare with standards. Used for ink differentiation and age determination. (**Marks:** Stepwise procedure 4 – Application in forensics – 2)
23. Secret writings use invisible inks or hidden impressions.
    Optical methods: UV light reveals fluorescent inks. IR photography detects carbon-based or heat-sensitive inks.
    Chemical methods: Iodine fumes, ninhydrin, or lemon juice reveal writings through color change.
    Other techniques: Oblique lighting and VSC help detect indentations or pressure marks, Used in espionage and fraud detection. **Marks:** Optical techniques – 3, Chemical techniques – 2 Application – 1
24. Paper examination identifies origin, type, and authenticity. Process includes: Physical examination – Size, color, texture, thickness, opacity. Optical examination – Watermarks, UV fluorescence, surface coating. Chemical analysis – Cellulose, fillers, coatings, pH tests. Instrumental analysis – IR spectroscopy or SEM for fiber content. Paper comparison assists in linking questioned and standard documents. (Marks: Steps explained – 4)

PART D – Essay Type Questions (Any 1 × 15 = 15 Marks)

25. Class characteristics in handwriting are those general features that are common to a group of writers who share a similar system of writing—such as people taught the same copy-book style, or trained in the same regional or institutional method. These features are not unique to an individual but reflect a common pattern of formation and design.

1. Definition and Nature

Class characteristics are *generic writing habits* learned during initial writing instruction.
They represent the framework of handwriting before individual traits emerge.
They help the examiner narrow down possible writers to a particular population group or educational system.

2. Examples of Class Characteristics

Slant and slope – Right or left inclination typical to a copy-book style.
Letter formations – Similar shapes of ‘a’, ‘g’, ‘k’, or ‘y’ found in certain cursive styles.
Connecting strokes – Whether letters are connected or separated in a word.
Pen lifts and line quality – Even, rhythmic strokes taught during schooling.
Spacing pattern – Uniform spacing between letters or words as practiced habitually.

3. Importance in Forensic Examination

Class traits assist in identifying the *writing system* used, e.g., Palmer, Zaner-Bloser, or Indian regional scripts.
They are crucial in early stages of comparison to group writings before focusing on individual traits.
Recognition of class traits prevents examiners from falsely attributing similarities caused by common instruction rather than authorship.

4. Distinction from Individual Characteristics

Individual characteristics evolve through repeated practice, physical changes, and personality.
While class traits identify the *group*, individual traits establish *identity*.
Both are evaluated together for conclusive opinion.

5. Forensic Application and Examples

In examining forged signatures, the forger often mimics class traits but fails to reproduce natural individual habits.
Class features also assist in identifying origin of anonymous writings (e.g., regional or institutional background).

(Definition and concept – 3 Examples with explanation – 4 Importance and distinction – 4 Forensic relevance and examples – 3 Conclusion and coherence – 1)

26. The examination of a forged cheque demands a systematic, multi-instrumental approach to detect alterations, additions, or erasures. The integration of several optical and analytical tools provides comprehensive results.
1. Role of Different Instruments
a) Light Sources (Visible, UV, and IR)
Various wavelengths reveal alterations invisible to the naked eye.
Oblique light detects indentations or surface disturbance.

UV light exposes fluorescent inks, erasures, and added entries.

IR absorption and luminescence distinguish inks of similar color but different composition.
b) Stereomicroscope
Provides three-dimensional visualization of strokes, erasures, and pen pressure.
Helps differentiate between original and added writings, as well as sequence of strokes (whether signature written before or after printing).
c) ESDA (Electrostatic Detection Apparatus)
Detects indented writings produced when a document is placed over another during writing.
In a forged cheque, ESDA can reveal impressions of genuine writings or prior attempts at falsification.
Non-destructive and preserves original evidence.
d) VSC (Video Spectral Comparator)
Employs multi-spectral imaging to examine inks, security features, watermarks, and background printing.
Identifies chemical differences between inks through spectral response.
Useful in detecting substitution of amounts or payee names.
e) Photomicrographs
High-resolution photographs taken through microscopes for documentation.
They record microscopic evidence of tampering, fiber disturbance, or ink deposition.
Essential for inclusion in forensic reports and court presentation.

2. Integrated Analytical Workflow

Visual and light source examination – initial observation for overt changes.
Microscopic analysis – confirmation of surface or ink layering.
ESDA scanning – recovery of latent impressions.
VSC testing – spectral differentiation of inks and security features.
Photomicrographic documentation – permanent visual record supporting expert opinion.

3. Forensic Interpretation- Combination of results determines whether writing is genuine or altered. Detects multiple ink entries, erased payee names, or forged signatures. Maintains chain of custody and scientific reliability of the report.

4. Case Relevance- In cheque fraud cases, these combined tools can expose insertion of extra digits, overwritten signatures, or replacement of payee names without damaging the document.

(Description of each instrument – 6 Workflow integration and analysis – 4 Forensic interpretation and examples – 3 Conclusion and clarity – 2)